

Supply *Module*

Spatial mapping of renewable energy plants, annual yield estimation via provincial capacity factors, and hourly temporal disaggregation of production into 24-element arrays — for all RES types in the region.

PostgreSQL 15

PostGIS 3

Solar · Wind · Hydro · Biomass

Regional scope: This documentation covers the **Sicilia** implementation (schema: `sicilia`). The identical SQL pipeline applies to **Puglia** and **Calabria** with schema substitution only.

FILE	LANGUAGE
Supply_Module_Sicilia/	SQL (PostgreSQL + PostGIS)
OUTPUT TABLE	RES TYPES
sicilia_actual_plants_hourly_production	Solar · Wind · Hydro · Biomass · Geothermal

Environment Optimisation & Spatial Mapping

1.1 Memory & Lock Management

SET work_mem = '1GB' and termination of idle user connections prevents deadlocks and ensures sufficient memory for the large array aggregation operations in step 4.

2.1 Primary Spatial Join — Plant Location to Province & Cabin

Each renewable plant point (sicilia_actual_plants.geom) is joined to its containing province boundary and primary cabin service area using **ST_Within()**. Results are stored in the mapping table sicilia_actual_plants_spatial_mapping.

```
INSERT INTO sicilia_actual_plants_spatial_mapping
SELECT a.fid, p.cod_prov, c.cod_ac
FROM sicilia_actual_plants a
LEFT JOIN prov_2024 p ON ST_Within(a.geom, p.geom)
LEFT JOIN primarycabins_2025 c ON ST_Within(a.geom, c.geom);
```

2.2 Topological Rescue for Orphaned Plants

Plants with NULL cod_prov or cod_ac — typically those located exactly on administrative boundaries — are rescued by assigning the **nearest province and cabin** using the <-> distance operator with LIMIT 1.

Annual Production Calculation

3.1 RES Source Classification

The Italian GSE source code (`ai_fonte`) is mapped to 5 standardised English type labels using `UPPER() CASE WHEN` logic:

<code>AI_FONTE (ITALIAN)</code>	<code>MAPPED TYPE</code>
<code>SOLARE</code>	<code>solar</code>
<code>EOLICA</code>	<code>wind</code>
<code>IDRAULICA</code>	<code>hydroelectricity</code>
<code>BIOGAS / BIOMASSE / RIFIUTI</code>	<code>biomass</code>
<code>GEOTERMICA</code>	<code>geothermal</code>

3.2 Yield Estimation via Utilisation Hours

Annual production is estimated as:

$$\text{yearly_production_kwh} = \text{potenza_kw} \times \text{utilisation_hours}$$

Where `utilisation_hours` is a provincial capacity factor sourced from `sicilia_actual_plants_utilization_hour` — a lookup table with one row per province per RES type, representing the equivalent full-load hours per year.

Hourly Production Disaggregation (Array-Based)

Annual production is distributed into 24-hour arrays using the same coefficient mechanism as the Demand Module — but with production-specific profiles per RES type.

4.1 UNLOGGED Output Table

The output table `sicilia_actual_plants_hourly_production` is created as **UNLOGGED** during the loop to eliminate WAL write overhead, dramatically improving insert performance. It is converted back to LOGGED after the loop completes.

4.2 Monthly PL/pgSQL Iteration

A `FOR m IN 1..12 LOOP` block processes one month at a time. Each iteration creates a temporary indexed lookup from the production profile table — filtered to month `m` — to minimise memory overhead.

```
CREATE TEMP TABLE temp_prod_lookup AS
SELECT cod_prov, type_res, fr_fs, hour, coefficient
FROM sicilia_actual_production_hourly_profile_lookup
WHERE month = m;
```

4.3 Cartesian Cross-Multiplication → 24h Array

Each plant's annual kWh is joined to all 24 hourly coefficients of its matching province and type. The product `yearly_production_kwh × coefficient` gives the absolute hourly production. `array_agg(... ORDER BY hour)` assembles the 24-element array.

```
p_k = yearly_production_kwh × coefficient
array_agg(p_k ORDER BY hour) → real[24]
```

4.4 **Finalisation & Indexing**

After the loop, the UNLOGGED table is converted to LOGGED with ALTER TABLE ... SET LOGGED. A B-tree index on plant_fid and ANALYZE are run to optimise downstream join performance in the Energy Balance module.

OUTPUT TABLE: SICILIA_ACTUAL_PLANTS_HOURLY_PRODUCTION

plant_fid	Plant identifier (int8)
month	Month number (1–12)
fr_fs	Day type: FR (working) or FS (festive/weekend)
type_res	RES type: solar / wind / hydroelectricity / biomass / geothermal
production_kwh_arr	24-element hourly production array (real[])



Note: The initial draft of this README documentation was generated with the assistance of Claude AI and subsequently reviewed, verified, and edited by the author for accuracy.